Validation of a sharp-interface void-bubble solver for long bubbles in circular tubes

Vatsal Sanjay

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Abstract

A long bubble translating through a wetting liquid in a circular tube deposits an annular film on the wall. We validate an axisymmetric sharp-interface finite-element solver for this problem in which only the liquid is meshed, the bubble interior is a true void of uniform pressure, and the steady state is obtained in a frame co-moving with the bubble rather than by marching a laboratory-frame transient. The bubble speed is a global unknown closed by the void centroid, not a regression against a tip trajectory. Twenty-five capillary numbers span $1.0 \times 10^{-4} \leq Ca_b \leq 14.8$, five decades. In the thin-film limit $H/Ca_b^{2/3}$ extrapolates to 1.3370, within 0.03–0.04% of Bretherton's $0.643\,3^{2/3} = 1.3375$, so the asymptotic coefficient is recovered rather than assumed. The dynamic pressure excess across the front meniscus independently recovers Bretherton's other coefficient, 3.58, to 0.4%. At finite capillary number the solver agrees with the published axisymmetric finite-element results of Giavedoni and Saita to +0.12% at $Ca_b = 1.0$, −0.18% at 2.0 and +0.05% at 10.0 (the last of those is printed text, not a digitised figure). It departs from the Aussillous–Quéré representation of Taylor's measurements by at most 1.37% anywhere below $Ca_b = 2$, with two sign changes, and by −4.56% at $Ca_b = 14.8$ where that fit is extrapolated far beyond its data. Seventeen verification stages are reported, including a displacement test showing the computed state is an attractor rather than merely a fixed point, a route-independence test showing the curve does not carry its own history, and an azimuthal linear-stability sweep in which the twenty-five states are stable to $k = 0, 1, 2$ and, except for one untested production row, to $k = 3$. We also find the rear cap ceasing to be convex between $Ca_b = 1.024$ and 1.048 in Stokes flow, a bracket containing the 1.026 obtained by extrapolating the independent Stokes correlation of Balestra and co-workers, and we resolve the vortex ring trapped in the resulting indentation. At the top of the sweep the Stokes film moves by 0.78% at $Re = 1.47$. A companion volume-of-fluid campaign lies above the present curve by up to 4.6% near $Ca_b = 1$; the cause of that difference is not established, though two candidates are eliminated.

1 Scope

This report validates one solver. It is a derived workstream of a volume-of-fluid campaign on the same physical problem, and the two are designed to be compared point by point; where that comparison appears below, it is labelled as such and no claim is made about which code is closer to the truth.

Three statements are worth separating at the outset, because they are established by different evidence and carry different weight. That the discrete equations are solved accurately is shown by the verification stages of Section 4, which compare against analytic answers, against independent solve routes, and against the solver's own refinement. That the equations being solved are the intended ones is shown by the limiting-case result of Section 5, which recovers a coefficient the solver was never given. That the answer is correct at finite capillary number rests on agreement with an independent published computation, in Section 6, and is therefore the weakest of the three.

2 Problem and reference laws

Let R_t be the tube radius, R_0 the volume-equivalent bubble radius, U_b the bubble speed and $\overline{U}$ the imposed mean liquid speed. With μ the liquid viscosity and σ the surface tension,

$$H = \frac{h_{\mathrm{flat}}}{R_t}, \qquad Ca_b = \frac{\mu U_b}{\sigma}, \qquad Ca_{\mathrm{in}} = \frac{\mu \overline{U}}{\sigma}. \tag{1}$$

The calculation prescribes Ca_{in}; Ca_b is an output. Every reference law is therefore evaluated at the computed bubble speed, never at the imposed one.

For a long bubble at vanishing capillary and Reynolds number, Bretherton (1961) matched the viscous dynamic meniscus to the static outer shape and obtained

$$H_B = 0.643\,(3Ca_b)^{2/3} = 1.3375\,Ca_b^{2/3}. \tag{2}$$

The rounded coefficient 1.34 is conventional and is used in the companion campaign's tables. The unrounded value is used throughout this report and in every figure, because the deviations discussed here are as small as 0.2% and the two conventions differ by about that much in the thin-film limit. Where the companion data are plotted, their deviations are recomputed from the raw H and Ca_b against Eq. (2), so that no part of any difference between the two codes is an artefact of the prefactor.

At finite capillary number Aussillous and Quéré (2000) represented the circular-tube measurements of Taylor (1961) by the Padé form

$$H_T = \frac{H_B}{1 + R\,H_B}, \qquad R = 2.5. \tag{3}$$

This is an empirical fit, not a solution of the governing equations, and its range of support is set by Taylor's data. Klaseboer et al. (2014) re-derived the same functional form from a matched spherical-cap construction and prefer $R = 2.79$. Equation (3) is used below as a reference curve that makes small differences visible, and not as a standard of correctness; Section 6 reports where it ceases to represent the computed film.

Bretherton's second observable follows from mass conservation. If the deposited film is at rest in the laboratory frame, then

$$\frac{Ca_b}{Ca_{\mathrm{in}}} = \frac{1}{(1-H)^2}, \qquad m = 1 - (1-H)^2, \tag{4}$$

with m the fraction of liquid left behind. Equation (4) is used here as a diagnostic, not as an input: the solver computes Ca_b, H and the fluxes independently, and the residual in Eq. (4) is reported rather than enforced. A film that is genuinely not stagnant is physics, not error.

3 Formulation

The solver computes axisymmetric Stokes or Navier–Stokes flow in the liquid alone, with a sharp free surface carrying surface tension. Three choices distinguish it from a laboratory-frame two-phase calculation of the same problem, and each is load-bearing.

The bubble interior is a void. No gas domain is meshed and no volume-of-fluid field exists. Surface tension acts on the liquid–void interface, whose nodes are mesh nodes, so the interface is not reconstructed from a colour function and the film thickness is read directly off the discretisation rather than inferred from it. The interior pressure is uniform.

The bubble volume is fixed and the void pressure is its Lagrange multiplier. The void pressure is not zero. The cap curvature of a confined bubble is set geometrically at roughly $2/(R_t - h)$, so the normal-stress balance determines the interior pressure from the local liquid pressure; changing the volume changes the bubble's length, not its cap curvature. With the interior pressure pinned instead and the volume left free, the computed volume would track the axial pressure drop and hence the window length, which is the very dependence the co-moving formulation exists to remove. Fixed volume is also the faithful analogue of an incompressible gas phase.

The frame co-moves with the bubble and its speed is an unknown. In a laboratory frame the domain must be long enough to contain the trajectory, and the steady state is a transient sampled when a stopping criterion fires. In the co-moving frame the bubble does not translate through the mesh; the window need only be long enough for the cap disturbance to decay to the fully developed profile, and the steady equations are solved directly. The frame speed U_b enters as a global degree of freedom closed by requiring the void's axial first moment to be stationary. Because the fully developed solution in a circular tube *is* Poiseuille flow, imposing it at both ends is the infinite-tube condition rather than an approximation to it.

The working nondimensionalisation is $\sigma = \mu = R_0 = 1$, so Ca_{in} is the imposed mean inlet speed and the Laplace number $La = \rho$. All results below are at $La = 0$, that is, Stokes flow, with $R_t/R_0 = 0.7$ and bubble volume $\frac{4}{3}\pi R_0^3$. Section 4 reports the measured size of the inertial correction at $La = 1$ on the thin-film state, and Section 8 quotes the Reynolds bound measured at the top of the sweep.

The implementation uses pyoomph (Diddens and Rocha, 2024), which is built on `oomph-lib` (Heil and Hazel, 2006) and GiNaC (Bauer et al., 2002), with Taylor–Hood elements for the flow and a hyperelastically smoothed moving mesh. Problem sizes range from 1.1×10^4 to 3.7×10^5 degrees of freedom.

4 Verification

Table 1 lists the staged checks. A stage earns its place by being able to fail: restating how well Newton converged is not a check, so every stage below compares against something the solver did not choose—an analytic answer, an independent route to the same state, a different discretisation, or a limit in which the answer is known.

Three of these deserve comment, because they test things a residual norm cannot.

Window independence is tested by the quantity that would expose it. Rather than refining the pad and observing that the answer stops moving, S5 compares the exact Poiseuille outlet with a traction-free one at the same pad. If the window is effectively infinite the outlet condition cannot matter, so this gap must collapse faster than either condition converges alone. At $Ca_{\mathrm{in}} = 0.05$ the relative gap in H is 1.29×10^{-3}, 5.1×10^{-6} and 3.0×10^{-11} at pads of one, two and five tube radii: eight orders against four for the individual convergence. Varying the pads separately shows that the *front* pad controls and the rear does not, by a factor of about 600 at pad one. This is the Landau–Levich geometry: in the bubble frame the wall moves rearward, so the advancing front meniscus is the coating meniscus that sets the film and the rear meniscus only receives it. A symmetric-only study would have concealed this.

Steadiness is tested by displacement, not by persistence. The reported states come from a stationary solve, which is an exact fixed point of the implicit time discretisation as well as of the steady equations. Marching from such a state therefore leaves it in place to round-off whether it is an attractor or a saddle, and persistence alone shows almost nothing. S9 instead displaces the state—to the converged solution at a capillary number 5% away, then restores

the parameter—and marches in time alone with no stationary polish. The imposed interface displacement of 4.6×10^{-4} to 3.6×10^{-3} in r/R_t decays to between 0.03% and 0.57% of itself within 1.5 film renewals, and H returns to within 1.4×10^{-5}.

The horizon of that test is quoted in film renewals L_b/U_b rather than in capillary times, and deliberately. That renewal time is 3172 time units at $Ca_b = 1.0 \times 10^{-3}$ and 0.51 at $Ca_b = 14.8$, a factor of six thousand across the sweep, so a single nondimensional horizon quoted uniformly would claim four orders of magnitude more evidence at one end than at the other.

The curve is not allowed to carry its own history. Points up to $Ca_{\mathrm{in}} = 0.7$ are each solved from their own seeded shape, independently of their neighbours; above that they come from an ascending continuation ladder, because a fresh transient costs a full relaxation per point where a continued step costs under a second. That switch is a matter of cost and not of capability: S10 measures a fresh solve succeeding as high as $Ca_{\mathrm{in}} = 2$, that is $Ca_b = 4.79$. Where both routes work they agree to 9.1×10^{-5} in Ca_b and 8.2×10^{-5} in H, and walking the ladder up to $Ca_{\mathrm{in}} = 6$ and back down returns the same states to 8×10^{-13}.

S10 also varies the bubble's length at fixed volume, by changing R_t/R_0 from 0.7 to 0.5. This lengthens the bubble by a factor of 1.8 and moves H by 1.3×10^{-3} at $Ca_b = 0.011$, falling to 1.9×10^{-4} at $Ca_b = 0.585$. The effect survives doubling the film resolution, which changes each confinement by about 10^{-5} and leaves the difference at 1.33×10^{-3}, so it is a resolved finite-bubble effect and not a discretisation error. It also bounds how much of any cross-code difference could be attributed to a mismatch in R_0/R_t.

Across the production sweep the void volume is conserved to at most 2.1×10^{-8} and the residual in the stagnant-film identity, Eq. (4), reaches at most 6.0×10^{-4}. The latter couples the frame speed, the film thickness and the flux in one relation and would fail if any of them were wrong; it is a consistency check on mass conservation and film flatness, and it is not by itself evidence that H is correct.

Azimuthal stability is labelled, not assumed. S12 showed that the azimuthal eigenproblem runs on this moving mesh once the pressure datum is the integral constraint rather than a nodal fixation. S13 marches the S12 state after an eigenfunction perturbation: the period of the rightmost $k = 0$ pair matches the eigenvalue to machine precision, and the fitted decay agrees to 14%. S14 then labels every committed sweep state for wavenumbers $k = 0, 1, 2$. The wavenumber is written k here because m is already the deposited-liquid fraction. All twenty-five production rows, and the eight states recomputed at doubled film resolution, are stable. There is no sign change to bracket. A pad-doubled window at $Ca_{\mathrm{in}} = 6$ agrees on the three labels and moves H by 5.3×10^{-4} relative. Growth rates are plotted as $\mathrm{Re}(\lambda)\, R_t/U_b$ in Figure 1, not as raw λ, because λ is in units of $\sigma/(\mu R_0)$ and its meaning changes across five decades. Zero is invariant; nothing in the figure is an extrapolation off the computed ladder. S16 checks that the pressure datum does not move that axisymmetric pencil. At $Ca_{\mathrm{in}} = 0.05$ the rightmost pair is $-0.123205 \pm 0.157796\,\mathrm{i}$ under nodal fixation with k unused, under the integral datum with k unused, and under the integral datum with $k = 0$. The three base states agree in Ca_b to 5.0×10^{-14}. Angular eigensolving remains refused under fixation; the class default is unchanged. S18 labels the same twenty-five states at $k = 3$. There is no sign change to bracket. Twenty-four production rows are stable. The production row at $Ca_{\mathrm{in}} = 5$ ($Ca_b = 12.30$) is untested: two shift-invert sets disagree on the imaginary part of the rightmost eigenvalue, while both remain damped. The doubled-film state at that station, and a pad-doubled window, are stable. Production rows at $Ca_{\mathrm{in}} = 4$, 3 and 2.5 failed on a downward climb from $Ca_{\mathrm{in}} = 6$ and were re-solved from the $Ca_{\mathrm{in}} = 0.7$ seed; those three are stable. Spurious-mode checks at $Ca_{\mathrm{in}} = 0.05$, 0.5 and 6 find no near-zero $k = 3$ eigenvalue.

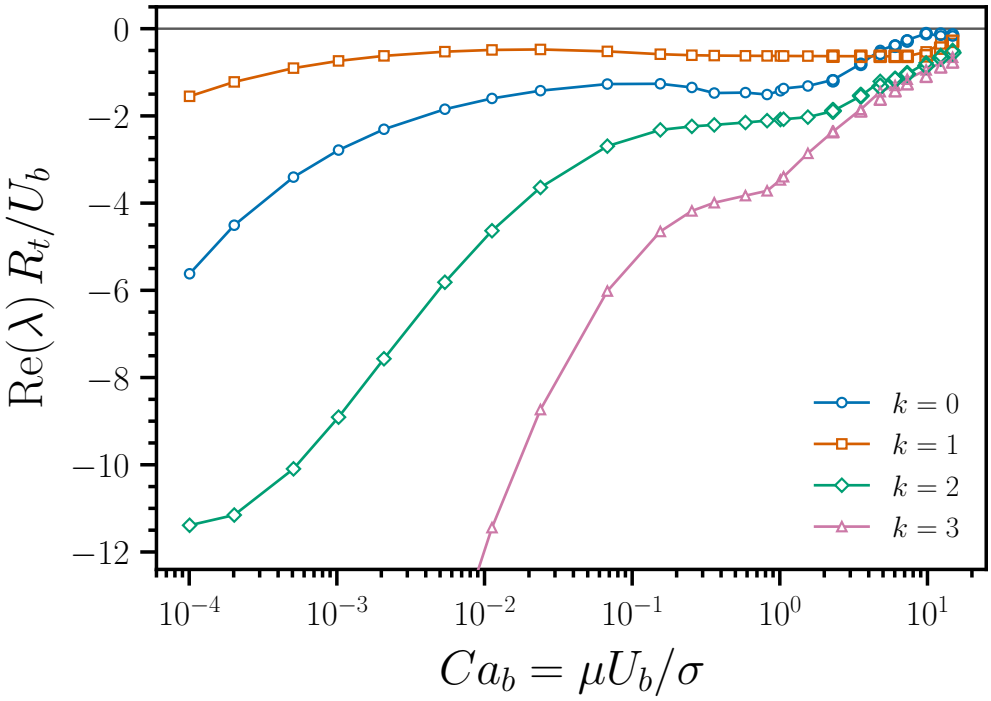


Figure 1: Scaled growth rate of the rightmost eigenvalue against Ca_b for azimuthal wavenumbers $k = 0, 1, 2, 3$. Filled markers are the production film resolution; open markers at $Ca_b \geq 2.3$ are the doubled-film check. The $k = 3$ production row at $Ca_b = 12.30$ is omitted (untested). At $Ca_b \lesssim 10^{-2}$ the $k = 3$ rate lies below the frame and remains negative. The tabulated data are supplied in `data/s14-azimuthal-growth.csv`.

5 The Bretherton limit

A line of slope 2/3 over a moderate range does not establish the asymptotic coefficient. The test is whether $H/Ca_b^{2/3}$ tends to 1.3375 as $Ca_b \to 0$, and at the smallest capillary number computed here the ratio is 1.3252, still 0.92% short. That shortfall is not a discrepancy: the leading confinement correction is itself of order $Ca_b^{2/3}$, so

$$\frac{H}{Ca_b^{2/3}} \simeq H'_B - R\, H'^2_B\, Ca_b^{2/3} + \cdots, \qquad H'_B = 1.3375, \tag{5}$$

and the ratio must be extrapolated rather than read off. Fitting Eq. (5) linearly to the two smallest points gives an intercept of 1.33697, which is 0.039% below Bretherton's coefficient; a quadratic fit to the five smallest gives 1.33707, or 0.032% below. The solver therefore recovers a number it was never given, to better than one part in two thousand.

A second coefficient, from a different observable. The film is not the only quantity Bretherton predicts, and agreement on one observable does not imply agreement on another. The pressure excess of the bubble over the liquid ahead of it is

$$\Delta p_{\mathrm{front}} = \frac{2\sigma}{R_t}\left[1 + 1.79\,(3Ca_b)^{2/3}\right] = \frac{2\sigma}{R_t} + \frac{3.58\,\sigma}{R_t}(3Ca_b)^{2/3}. \tag{6}$$

The coefficient 3.58 is the *leading-meniscus* correction quoted in Bretherton's abstract; it is not the 4.52 that applies across a complete long bubble, front and rear together, and the two must not be interchanged. The quantity computed here is the front-meniscus excess: the void pressure less the liquid pressure at the front tip station, obtained by removing the Poiseuille drop between

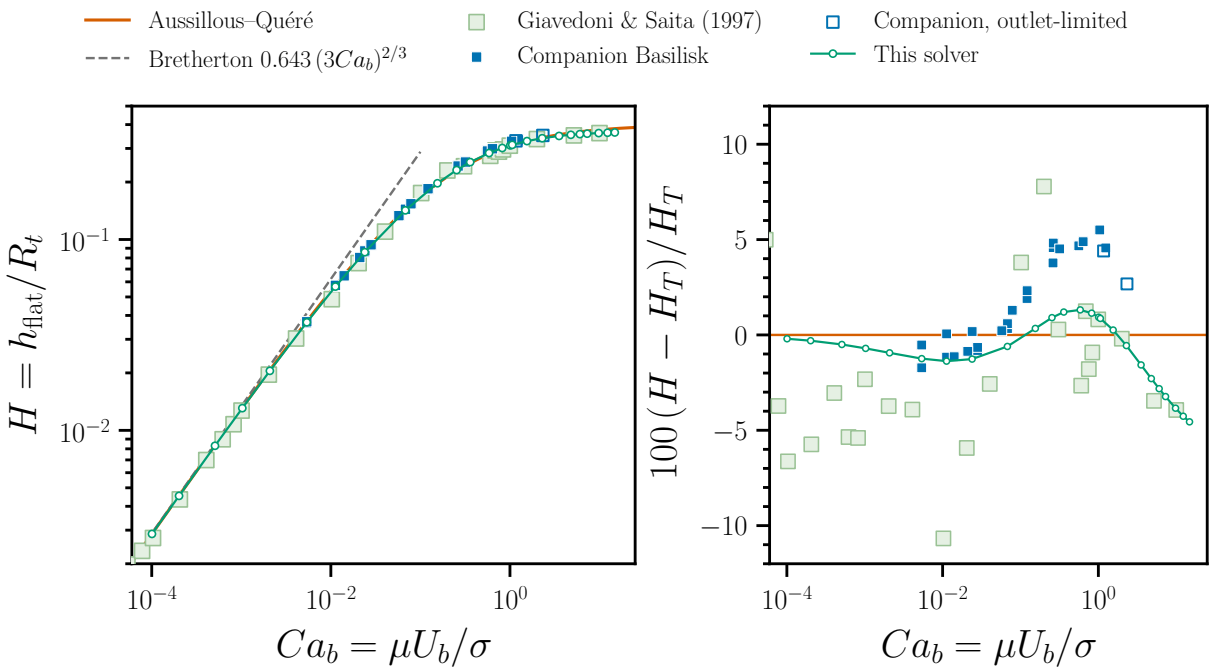


Figure 2: Deposited central film against the computed bubble capillary number. Left: H with the Bretherton asymptote and the Aussillous–Quéré fit. Right: the deviation of every series from that fit, all recomputed from raw H and Ca_b against the unrounded coefficient of Eq. (2). Most literature points are digitised from figures and are drawn as large soft symbols for that reason; their scatter at small Ca is reading error, not disagreement. The points at $Ca = 2$ and $Ca = 10$ are printed values from the same paper. Two companion cases reached the end of their laboratory-frame domain and are drawn open.

that station and the outlet, which is window-dependent by construction and has nothing to do with the bubble. Forming $(\Delta p_{\mathrm{front}} - 2\sigma/R_t)\, R_t/(3Ca_b)^{2/3}$ and extrapolating as before gives 3.594 from the two smallest points and 3.569 from the three smallest, bracketing 3.58 within 0.4%. This is an independent limiting-case check: it constrains the normal-stress balance and the void pressure, neither of which the film thickness alone tests.

The same fit returns the coefficient of the first finite-capillary-number correction. The initial slope corresponds to $R = 2.94$ from the three smallest points, drifting to $R = 2.72$ when five are used. This is to be compared with $R = 2.5$ in the Aussillous–Quéré fit and $R = 2.79$ preferred by Klaseboer et al. (2014). The measured thin-film value sits above both. Fitting Eq. (3) with R free over the whole sweep gives $R = 2.67$ with a root-mean-square residual of 3.6%, so no single value of R represents five decades; this is the quantitative form of the statement that Eq. (3) is an interpolation over a finite range.

6 The deposited film at finite capillary number

Figure 2 shows the computed film over five decades against the reference laws, the digitised axisymmetric finite-element results of Giavedoni and Saita (1997), and the companion volume-of-fluid campaign. The tabulated data are supplied in `data/film-sweep.csv`.

Departure from the empirical fit. The deviation from Eq. (3) is −0.20% at $Ca_b = 1.0\times10^{-4}$ and shrinks monotonically towards zero as Ca_b falls, as it must if Section 5 is right. Going up in capillary number it reaches a minimum of −1.37% at $Ca_b = 0.011$, changes sign near $Ca_b = 0.115$, reaches a maximum of +1.31% at $Ca_b = 0.585$, changes sign again near $Ca_b = 1.75$, and then falls

monotonically to -4.56% at $Ca_b = 14.8$. Nowhere below $Ca_b = 2$ does it exceed 1.37%. Mesh convergence is 7.9×10^{-5}, two orders below this structure, so the two sign changes are resolved features of the solution and not numerical noise. Their interpretation is that a one-parameter Padé fit cannot follow the true curve over five decades, which is the same conclusion the free-R fit of Section 5 reaches by another route. We do not interpret the -4.56% at $Ca_b = 14.8$ as an error in either the solver or the fit: it is the fit being extrapolated an order of magnitude beyond the data it represents.

Agreement with an independent computation. The closest available independent check is Giavedoni and Saita (1997): the same geometry, the same formulation class, a different implementation, and published. Twenty-two of their twenty-four points fall inside the present range; $Ca = 2$ and $Ca = 10$ are printed ($m = 0.559$ and $m = 0.592$, $h_\infty = 0.361$ on p. 2424), the rest digitised from figures. Interpolating the present sweep onto them, the difference is $+0.12\%$ at $Ca = 1.0$, -0.18% at 2.0, $+1.00\%$ at 5.18 and $+0.05\%$ at 10.0; over the nine points with $Ca \geq 0.3$ the mean difference is $+1.16\%$ with a root-mean-square of 1.84%. Below $Ca \approx 0.05$ the differences widen to several per cent, which is the expected consequence of reading a thin film off a logarithmic figure: a fixed pixel error becomes a large relative error as the film thins. This agreement is the single strongest piece of evidence in this report that the finite-capillary-number film is right, and it should be read as consistency across several independent points rather than as a calibration.

Comparison with the companion campaign. The companion volume-of-fluid calculations lie above the present curve, by a mean of $+0.71\%$ below $Ca_b = 0.05$, $+2.47\%$ between 0.05 and 0.3, and $+3.85\%$ above 0.3, reaching $+4.6\%$ near $Ca_b = 1$. The two curves therefore agree in the thin-film limit and separate as the film thickens, and over the range where the literature points are reliable the present curve sits with them. The cause of the difference is *not established*. Two candidates have been eliminated. Inertia is not responsible: S8 measures the film moving by -0.0255% between $La = 0$ and the companion's $La = 1$ at $Ca_b = 0.068$, and S15 finds at most 0.78% anywhere with $Re < 1.47$, both far below the gap. A mismatch in bubble length is not responsible: S10 bounds its effect at 1.3×10^{-3}. Two candidates remain, and they are distinguishable by calculations that do not exist yet. The companion's bubble is a viscous gas with $\mu_g/\mu_c = 10^{-2}$ where the present interior is a true void, which is a genuine physical difference whose size is unknown. The companion also has no completed spatial refinement sequence, and its stopping rule accepts a film after an advance of order one bubble radius, which is a fraction of one film renewal at these capillary numbers. Settling this requires a refinement sequence and a longer-running case on the volume-of-fluid side; nothing on the present side can substitute for either.

7 The rear cap

Figure 3 shows converged bubble profiles at one capillary number per decade, and the indentation of the rear cap against Ca_b. At low capillary number the rear cap is convex and carries the familiar capillary ripple where it joins the film. Above $Ca_b \approx 1$ the cap ceases to be convex: the on-axis rear tip moves *ahead* of the rearmost point of the interface, so that the bubble's trailing surface folds back around an axisymmetric indentation. We measure that indentation as $\delta_{\mathrm{tail}} = z_{\mathrm{tip}} - \min z$ over the rear cap.

The onset is bracketed, not resolved. At both film resolutions the cap is strictly convex up to $Ca_b = 1.024$, and both give a non-zero indentation from $Ca_b = 1.048$ onwards, so the onset lies in $1.024 < Ca_b < 1.048$. It cannot be located more sharply than that: through $Ca_b \approx 1.1$ the two resolutions disagree by factors of four to forty, which is the signature of an indentation

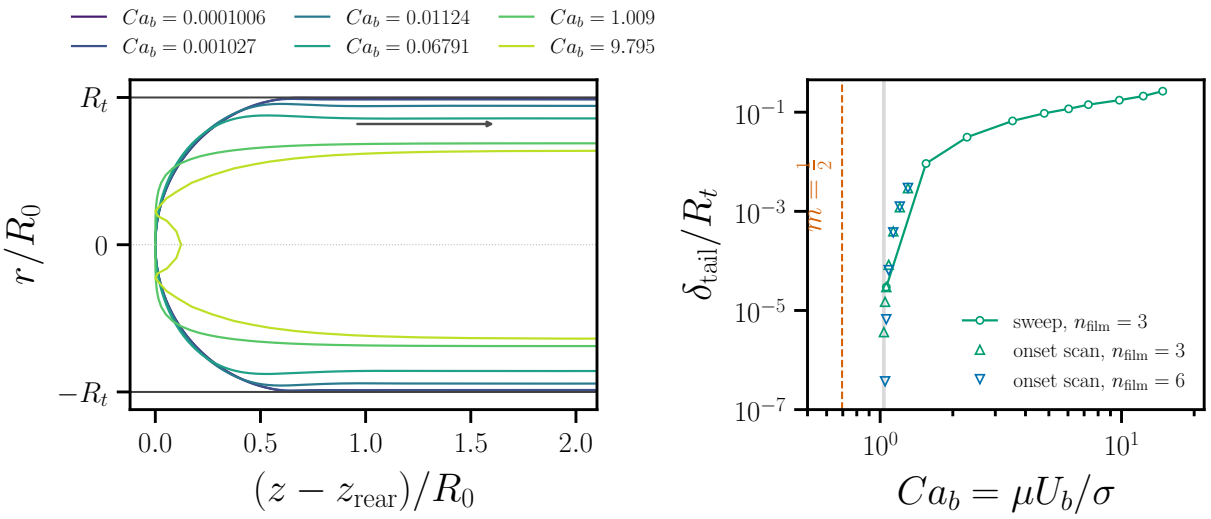


Figure 3: Left: converged interfaces, one per decade, each shifted to place its rearmost point at the origin and mirrored about the axis; the arrow gives the direction of bubble motion. Right: the rear-cap indentation, at two film resolutions. Below the shaded band the cap is strictly convex at both resolutions and the indentation is identically zero; within the band the two resolutions disagree by factors of four to forty, so the band is where the cap changes character and not a resolved onset. The dashed line marks Taylor's $m = \frac{1}{2}$ condition, which this solver places at $Ca_b = 0.695$ and which is a distinct and earlier transition.

at the level of the discretisation rather than of the solution. Once δ_{tail} exceeds about $4 \times 10^{-4} R_t$, at $Ca_b \gtrsim 1.1$, the two resolutions agree to 5% and to 3% beyond that. The indentation then grows monotonically to $0.27 R_t$ at $Ca_b = 14.8$. All of this is at $La = 0$, so inertia plays no part in it.

It is a separate transition from Taylor's. Far from the bubble the axial velocity on the axis in the bubble frame is $2\overline{U} - U_b$, which changes sign when $Ca_b/Ca_{\mathrm{in}} = 2$, that is when $m = \frac{1}{2}$: below it the liquid on the axis outruns the bubble and recirculates ahead of the nose, above it every streamline moves rearward in the bubble frame and the flow is complete bypass. Using Eq. (4), that condition is $H = 1 - 1/\sqrt{2} = 0.2929$, and the present sweep crosses it at $Ca_b = 0.695$. The rear cap is still convex there and remains so for a further 50% in capillary number. The two transitions are therefore ordered and distinct, and the bypass condition is not sufficient for the indentation. The computed m rises to 0.595 at $Ca_b = 14.8$.

This reproduces a published Stokes-flow prediction. A re-entrant rear cap in a tube at vanishing Reynolds number is not new, and the critical capillary number is not new either. Martinez and Udell (1990) predicted the onset of a re-entrant cavity at the trailing end of a drop at $Ca \approx 0.75$ by boundary integrals; Olbricht and Kung (1992) observed an indentation that entrains outer liquid for low viscosity ratio; Lac and Sherwood (2009) and Nath et al. (2017) find the same feature, the former reporting eventual breakup through a rear re-entrant jet.

The sharpest comparison is Balestra et al. (2018), who solve the axisymmetric Stokes equations for a confined droplet and fit the plane curvature of the rear cap as

$$\kappa_r R = \frac{1 + T_r (3Ca_b)^{2/3}}{1 + Z_r (3Ca_b)^{2/3}}, \tag{7}$$

with the capillary number built on the *droplet* velocity, as ours is. At viscosity ratio zero their tabulated coefficients give $T_r = -0.4725$, so the rear curvature changes sign where $(3Ca_b)^{2/3} = 1/0.4725$, that is at $Ca_b = 1.026$. Our bracket, $1.024 < Ca_b < 1.048$, contains that value.

Three things limit how much weight that agreement carries, and they should be read before it is quoted. The root is obtained by extrapolating their correlation to the edge of, and arguably just beyond, the capillary-number range over which it was fitted, so it is not a value they compute and report. It is derived from their tabulated coefficients rather than printed by them, and therefore inherits whatever error the fit carries. And their droplet volume, $12.9R^3$ against our $12.2R_t^3$, is close but not identical. What can fairly be said is this: two independent Stokes solvers, neither tuned to this transition, place it within about one per cent of each other. That is consistency, and it is the right weight to give it.

The same source predicts the other transition. Their $\lambda = 0$ film law with $Q = 2.483$ reaches $H = 1 - 1/\sqrt{2}$ at $Ca_b = 0.720$, against the 0.695 measured here, a difference of 3.4%. The ordering of the two transitions is worth recording, because Balestra et al. (2018) state in prose that the rear recirculation detaches at a film thickness *smaller* than the complete-bypass value, while their own $\lambda = 0$ fit puts curvature reversal at $H = 0.311$ and bypass at $H = 0.293$, the other way round. Our two independent measurements, 0.695 then 1.024–1.048, agree with their model and not with their prose.

What Giavedoni and Saita do and do not contain. We cite Giavedoni and Saita (1997) for the film data of Section 6, and it is worth being precise about what else that pair of papers supports. The 1997 calculation is of a semi-infinite gas finger with a single advancing meniscus, so it has no rear cap to invert; the recirculating pattern its abstract reports as overlooked is therefore not a rear-cap phenomenon. Giavedoni and Saita (1999) adds the trailing meniscus, and does report a rear-meniscus curvature change — but at finite Reynolds number, with the critical capillary number falling as inertia grows. We therefore exclude that pair as precedent for the *inertialess* inversion reported here, and not as precedent for the phenomenon at large.

What is not already published. Two things, both modest. We have found no published measurement of the indentation *depth* against capillary number, in any geometry; the monotone growth to $0.27R_t$ at $Ca_b = 14.8$ in Figure 3 appears to be new. And the steady branch is continued an order of magnitude beyond where the prior studies stop: they terminate near onset because their bubbles break up, whereas a fixed-volume void in a steady co-moving formulation cannot pinch off. That is a capability and also a caveat, stated in Section 8: this solver will continue a branch that a physical bubble may have left by breaking up, and nothing here establishes that these states are stable.

A vortex ring sits in the cavity. Figure 4 shows the flow around the cap in the frame of the bubble, which is the only frame in which the question is well posed: in the laboratory frame the liquid near the bubble simply follows the flow, whereas in the bubble frame the far field behind the cap is Poiseuille flow minus U_b, which is directed rearward at *every* radius. Any forward-moving liquid there is therefore a local structure and not the far field intruding.

There is such a region, and it is closed. At $Ca_b = 9.80$ the liquid inside the cavity moves forward along the axis, turns outward beneath the overhanging lip and rejoins the rearward stream: an axisymmetric vortex ring seated in the indentation, bounded by a dividing streamline. It occupies $-0.144 \le (z - z_{\mathrm{tip}})/R_0 \le -0.024$ and reaches to $r = 0.19R_t$. It is slow. The peak forward speed is 0.065 against a far-field on-axis value of -1.53 and a film speed near -9, so the eddy turns over at about four per cent of the through-flow. That weakness is why it is shaded in the figure rather than left to the streamlines to convey.

One mechanism was proposed and measured to be wrong. A natural explanation is that the film flux, which must be re-injected into the bulk behind the bubble, converges radially and stagnates on the axis, raising the pressure there and pushing the interface in. That is testable, because it predicts the sign of the radial pressure difference. Measuring it requires care: the

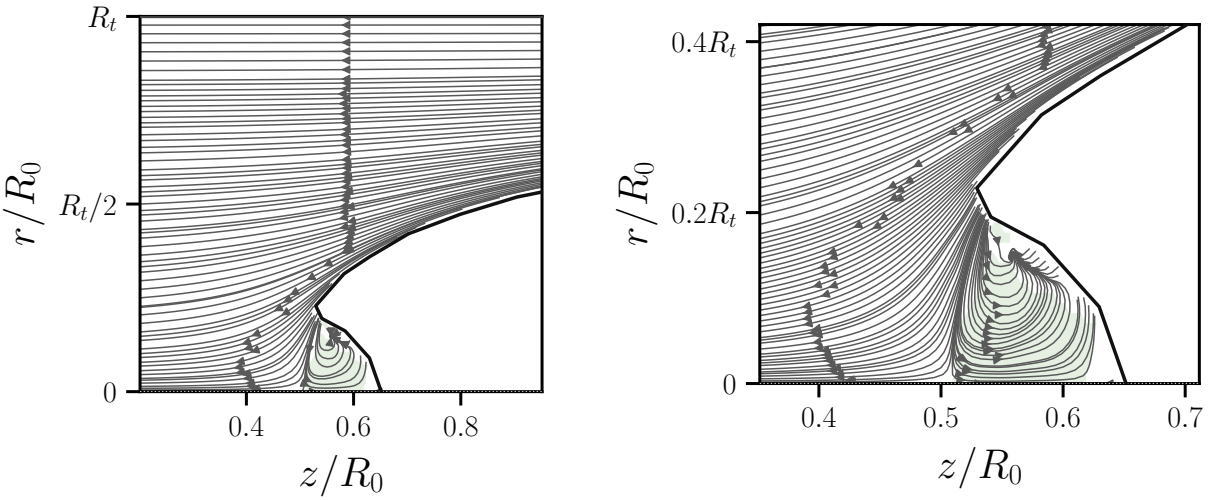


Figure 4: Streamlines about the re-entrant rear cap at $Ca_b = 9.80$, in the frame of the bubble, with the liquid shaded where it moves forward. Left: the cap, the deposited film and the wake. Right: the cavity, where the closed vortex ring is resolved. The heavy black curve is the interface and the heavy line at the top of the left panel is the tube wall, which runs rearward at $-U_b$ in this frame. Both panels are one converged solution, the sweep row at $Ca_{\mathrm{in}} = 4$.

axial gradient behind the bubble is about $8Ca_{\mathrm{in}}/R_t^2$ per unit length, of order 67 at $Ca_b = 10$, so sampling the axis and the wall at stations differing by even 0.02 manufactures a difference of 1.3 from nothing. With the axial trend removed by a local fit in each radial group, and restricting to stations strictly behind the rearmost point of the interface, the axis is found to be *below* the near-wall liquid in pressure, by about 9 at $0.25R_0$ behind the tip, decaying to 0.2 by $0.6R_0$ behind. The sign is opposite to the proposed mechanism, which is therefore rejected. What the measurement does show is that the tail sits in a strongly non-developed wake: at $0.5R_0$ behind the tip the annular liquid moves at -9.1 in the bubble frame while the core moves at -1.7, relaxing to the fully developed -1.83 on the axis within about $0.6R_0$.

A sufficient explanation must also account for the magnitude, and the scaling gives only a warning rather than an answer. Ca_b compares viscous with capillary stress on the global scales U_b and R_t, so at $Ca_b \simeq 10$ there is no reason to expect the deviatoric normal stress on the interface to be a small correction to the capillary term. That is an expectation about magnitudes and not a measurement: the local deviatoric stress depends on the velocity gradient at the interface, which in a cavity where the liquid barely moves need not follow the global scale at all. Settling the mechanism requires the full normal-stress balance evaluated along the interface, deviatoric part included, which this report does not attempt. Balestra et al. (2018) had already established that the viscous normal stresses at the caps cannot be neglected for $Ca_b > 10^{-3}$, so that framing is theirs and not ours; what is added here is the measurement along the cap rather than at the vertex, an order of magnitude beyond their range.

A published candidate mechanism exists, and we record it as a candidate rather than adopt it. Cappello et al. (2026) argue that a convergent stagnation point on the interface corresponds to a locally pushing normal stress and a divergent one to a locally pulling stress, so the inhomogeneous normal stress set up by the recirculating flow lowers the curvature near convergent points; on that reading the rear on-axis stagnation point is the convergent one, which would be why the rear cap and not the front inverts. It is consistent with the recirculation we resolve, and we have *not* tested it: doing so means measuring the stagnation-point structure on the interface and the normal stress there, which is the same unattempted calculation as above. The same paper states that rear recirculation in this regime "appears as soon as there is an invagination at the back of the bubble", with the supporting figure not shown; Figure 4 is that figure for a true void at

$Ca_b = 9.80$.

Two cautions for anyone reading the coincidence of numbers. Bubbles in viscoelastic liquids develop a rear cusp above a critical capillary number itself of order one, also at vanishing Reynolds number; that is a constitutive effect with a negative wake and a hysteretic dependence on bubble volume, and none of those features is present here. And the regime our high-Ca_b points occupy, $U_b/\overline{U} > 2$, is the one the recent literature calls *super-fast*; at $Ca_b = 9.80$ the present solution has $U_b/\overline{U} = 2.45$.

8 What is established, and what is not

The following are established by the evidence above. The solver reproduces Poiseuille flow, the static Laplace pressure, and window and mesh independence at the stated tolerances. The computed state is an attractor under time evolution, not merely a fixed point. Its curve does not depend on the route taken to compute it. Every reported Stokes state is linearly stable to azimuthal wavenumbers $k = 0, 1, 2$, and to $k = 3$ except for one untested production row at $Ca_{\mathrm{in}} = 5$. In the thin-film limit it recovers Bretherton's film coefficient to 0.03–0.04% and his front-meniscus pressure coefficient to 0.4%, both by extrapolation and neither given to the solver. Over $0.3 \le Ca_b \le 10$ it agrees with an independent published axisymmetric computation to about one per cent, and at three separate capillary numbers to better than 0.2%.

The following are not established. The film is validated through $Ca = 10$ by that printed Giavedoni and Saita point. Beyond it the $Ca_b = 14.8$ film is an unvalidated continuation of the same branch: there is no printed table and no second independent Stokes computation in the bibliography at $Ca_b > 10$. The departure from the Aussillous–Quéré fit there is attributed to extrapolation of the fit, and that attribution is an inference. The mechanism of the re-entrant rear cap is open, with one candidate measured and rejected. The difference from the companion volume-of-fluid campaign is resolved on the present side but its cause is unidentified; of the four candidates considered, two are eliminated and the remaining two require work on the companion side. Inertial effects are no longer a single-state caveat: S15 measures $\mathrm{d}H/\mathrm{d}Re$ at $Ca_b = 0.068$, 1.06 and 14.8, and the Stokes baseline holds to 0.78% for $Re < 1.47$. Continuation in Ca_{in} at $La = 1$ terminates near $Ca_{\mathrm{in}} = 2.91$ both with automatic remeshing and without it: S17 seeds that row with remesh, then climbs with remesh suppressed, holds 11104 degrees of freedom from $Ca_{\mathrm{in}} = 2$ to 2.9, and dies at the same $Ca_{\mathrm{in}} = 2.90851$ as S15. A remesh-off transient seed at $Ca_{\mathrm{in}} = 2$ fails before a state is obtained, so remesh is required to start the row but does not remove the barrier. The top-of-sweep inertial point is accordingly quoted at $La = 0.1$ rather than $La = 1$. S19 fills the named (Ca_{in}, La) grid. At $Ca_{\mathrm{in}} = 0.05$ the 3×3 and 6×6 pads agree in H to 7×10^{-8} at $La = 0.1$, 1 and 10; $La = 10$ there thins the film by 0.253% at $Re = 0.679$. At $Ca_{\mathrm{in}} = 0.5$, $La = 10$ the film is 5.8% thinner than Stokes ($H = 0.295$, $Ca_b = 1.006$, $Re = 10.1$), so inertia is no longer a perturbation at that station. The last completed $La = 1$ station, $Ca_{\mathrm{in}} = 2.9$, has $H = 0.346$ and $Ca_b = 6.77$. Allowed failures remain at $La = 10$, $Ca_{\mathrm{in}} = 2$ and at the $Ca_{\mathrm{in}} = 6$, $La = 0.1$ pad-doubled window. The S15 and S19 rows are not near the rear-meniscus threshold of Magnini and Herrada (2025): their modified Weber number is $We_p = We(2 - U_b/U_l)^2/2$ with $We = 2\rho U_l^2 R_t/\sigma$, which on the present scales is $0.7\, La\, Ca_{\mathrm{in}}^2(2 - Ca_b/Ca_{\mathrm{in}})^2$. Converted, the three S15 stations give $We_p = 7.2 \times 10^{-4}$, 2.5×10^{-3} and 0.55, all below $We_p^* \approx 3.65$ and the last two outside their Ca_l window; the S19 station at $La = 10$, $Ca_{\mathrm{in}} = 0.5$ has $We_p = 2.5 \times 10^{-4}$, so that 5.8% thinning is a Reynolds-number effect, not their rear threshold. At $Ca_{\mathrm{in}} = 2.9$, $La = 1$ one has $We_p = 0.664$. Our $We = La\, Ca_b^2$ is a different group; at the top of the sweep it is 21.5. Nothing here addresses drops, for which a true void cannot stand in for the interior phase and a second meshed phase is required.

The twenty-five Stokes states have been labelled for azimuthal wavenumbers $k = 0, 1, 2$ and $k = 3$. All labelled rows are stable; S14 and S18 report no crossing to bracket. The production row at $Ca_{\mathrm{in}} = 5$ remains untested at $k = 3$. The eigenproblem cannot distinguish breakup from

another axisymmetric state, and the labels are not a bifurcation study.

Data

All quantities plotted or tabulated in this report are supplied as compact CSV files under `data/`, and every figure is regenerated from them by the scripts in `figures/` without a solver or a simulation dump. The digitised points of Giavedoni and Saita (1997) record the source figure and the coordinates read; printed values record the page. The companion volume-of-fluid table is reproduced with that campaign's agreement and carries its own resolution measure and its outlet-limited flag, both of which are honoured in the figures.

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Table 1: Verification stages. Each is a script that exits non-zero on failure and writes a machine-readable receipt. Numbering has gaps because three intermediate stages were folded into the production path during development.

Stage	Question	Result
S0	Does the bare window reproduce Poiseuille flow?	Flux to 4.4×10^{-16}, pressure drop to 5.0×10^{-14}, worst of four configurations
S1	Does an unconfined void relax to a sphere at the Laplace pressure?	Radius to 1.1×10^{-5}, interior pressure to 2.2×10^{-6}
S5	Is the answer independent of the window length?	See below
S6	Is it independent of the mesh?	H moves by at most 7.9×10^{-5} relative
S7	Does the pressure datum affect any observable?	Two mechanisms agree to 2.6×10^{-14}
S8	How large is the inertial correction?	-0.0255% in H between $La = 0$ and $La = 1$ at one Ca
S9	Is the computed state an attractor?	Displacements decay to 0.03–0.57% in 1.5 film renewals
S10	Does the answer depend on the solve route?	Routes agree to 9.1×10^{-5}; retrace to 8×10^{-13}
S11	Where does the rear cap stop being convex?	Onset in $1.024 < Ca_b < 1.048$; resolutions agree above $4 \times 10^{-4} R_t$
S12	Is azimuthal linear stability available for this formulation?	Yes, with an integral pressure datum; $k = 0$ needs eight eigenvalues
S13	Does the rightmost eigenvalue match an independent time march?	Period to machine precision; decay rate to 14%
S14	Are the twenty-five states stable to $k = 0, 1, 2$?	All stable; no crossing; pad-doubled labels agree
S15	To what Re may the Stokes baseline be quoted?	Holds to 0.78% for $Re < 1.47$
S16	Does the pressure datum move the axisymmetric spectrum?	Rightmost λ agrees; Ca_b spread 5.0×10^{-14}
S17	Does $La = 1$ continuation past $Ca_{\mathrm{in}} = 2.91$ survive with remesh off?	No; it dies at 2.90851 with the mesh unchanged. Continuation limit
S18	Are the twenty-five states stable to $k = 3$?	No crossing; 24 of 25 production rows stable; $Ca_{\mathrm{in}} = 5$ untested
S19	What is H on the named (Ca_{in}, La) grid?	Stokes quote unchanged for $Re < 1.47$; $La = 10$ at $Ca_{\mathrm{in}} = 0.5$ thins H by 5.8%